
Lightweight Knowledge-Grounded Detection of Hateful Memes

Alachkar Joelle Bollack Alexis Tarabay Andrea

Group 03

Abstract

Hateful memes are hard to detect because the hateful meaning often depends on the interaction between image, text, humor, and cultural references. This becomes harder in multilingual memes, where language- and culture-specific cues may not be captured by general models. At the same time, large vision-language models are costly to store and deploy. We address this by using a lightweight MemeBLIP2-based student and adding external knowledge retrieval and MemeLens teacher distillation to provide additional contextual information. Knowledge retrieval helps more on MUTE than on FHM, suggesting that targeted Bengali lexical cues better match MUTE’s offensive patterns, while chat-based distillation provides only limited additional gains.

Keywords: hateful memes, multimodal learning, knowledge grounding, distillation.

1. Introduction

Hateful memes are challenging to detect because their harmful meaning often emerges from the interaction between image, text, humor, symbols, stereotypes, and cultural context, rather than from either modality alone [1]. This makes hateful meme detection a multimodal task where text-only or image-only models may miss the intended message.

The challenge increases in multilingual settings such as Bangla, where memes may include code-mixing, local expressions, and community-specific references that English-centric models struggle to capture. While large vision-language models (VLMs) offer stronger contextual reasoning, they are costly to store and deploy. This motivates the need for lightweight models that retain contextual information while reducing storage and inference costs.

Our objective is to test whether external knowledge and teacher supervision can improve a lightweight multimodal hate detector. Our model follows a MemeBLIP2-style de-

sign, freezing large pretrained encoders while training only lightweight projection, adaptation, fusion, and classification layers [2]. We compare three variants: an image-text student, a knowledge-augmented student using cross-attention, and a knowledge-aware student distilled from MemeLens-VLM [3]. Models are trained on English FHM and then transfer fine-tuned on Bengali MUTE.

We **hypothesize** that external, hate-related knowledge helps the lightweight model **recover missing contextual cues**, especially in language-specific settings. We also expect MemeLens distillation to **transfer useful reasoning** from a larger meme-specialized VLM to the student without increasing inference cost.

2. Related Work

The Facebook Hateful Memes dataset established hateful meme detection as a multimodal task, where image and text may appear harmless separately but become hateful together [1]. Multilingual benchmarks such as Multi3Hate further show that this task is language, culture, and community-dependent, and that current VLMs still struggle in multilingual and culturally grounded settings [3, 4].

Another line of work addresses the inefficiency of large VLMs. GatedCLIP and MemeBLIP2 reduce deployment cost by freezing large pretrained encoders and training only lightweight projection, fusion, adaptation, and classification modules [2, 5]. However, such lightweight models may lack the contextual reasoning and implicit background knowledge needed for culturally specific memes. Knowledge-grounded methods address this by injecting additional context: TRACE uses generated captions and cross-attention, while CROSS-ALIGN+ retrieves cultural and hate-related knowledge and fuses it through cross-attention [6, 7]. In parallel, knowledge distillation transfers information from larger teachers to smaller students through soft labels [8]; MemeLens provides a multilingual meme-specialized VLM that can serve as such a teacher [3].

Together, these works motivate our study of whether a **deployable MemeBLIP2-style detector** can benefit from **external knowledge grounding** and **MemeLens-based teacher supervision**.

3. Method

We compare **three configurations** built on the **same** underlying multimodal architecture.

Base The first variant is based on MemeBLIP2: images are encoded with a frozen BLIP-2 vision encoder, and meme text with frozen CLIP text embeddings. The image and text features are projected into a shared hidden space, passed through lightweight adapters, fused, and classified with an MLP. Only the projection, adapter, fusion, and classification layers are trained.

Context The second variant adds external knowledge retrieval. For each meme, we generate an image caption with BLIP and build a retrieval query from the caption and meme text. Candidate knowledge entries are ranked using BM25 lexical matching, MiniLM sentence-embedding similarity, and source scores. MiniLM is a compact Sentence-BERT model pretrained for sentence-level semantic similarity, which helps retrieve contextually related entries beyond exact word overlap. For FHM, retrieval uses GenericsKB together with English hate/offensive lexicons derived from HurtLex and MOL [9, 10, 11]. These resources provide categorized offensive cues such as insults, sexism, racism, and xenophobia. For MUTE, retrieval is restricted to Bengali HurtLex to better match Bangla offensive cues. The top five entries are represented by MiniLM embeddings and projected into the model hidden dimension. The meme’s image and text tokens query this knowledge through multi-head cross-attention, and a residual gate controls how much of the knowledge update is added before classification.

Context + Distillation The third variant builds on the external-knowledge model and adds chat-based teacher distillation. We use MemeLens-VLM as the teacher because it is a Qwen3-VL-8B-based model fine-tuned for multilingual meme understanding. The teacher receives the image and meme text, then returns a predicted label and confidence score, which are parsed into soft class probabilities. For FHM, we also include retrieved lexicon-based context in the teacher prompt, since this improved MemeLens predictions during validation. For MUTE, we omit this context because English-style prompting with retrieved Bengali lexicon entries did not improve the teacher, likely due to the language mismatch between prompt and context. The training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(y, p_s) + \lambda_{\text{KD}} \mathcal{L}_{\text{KL}}(p_t^\tau \| p_s^\tau)$$

where p_s is the model prediction, p_t is the teacher distribution, $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss with the true label, and $\mathcal{L}_{\text{KL}}$ encourages the model to match the teacher’s softened probabilities. The temperature τ controls the softness of the distributions, and λ_{KD} controls the strength of the teacher signal.

Training is performed in two stages: we first train the three variants on English FHM, then use the corresponding check-

points to initialize transfer fine-tuning on Bengali MUTE. FHM contains 5,552/1,190/1,190 train/validation/test examples after balancing, with 3,966 hateful and 3,966 non-hateful memes. MUTE contains 2,222/475/475 examples, with 1,586 hateful and 1,586 non-hateful memes. This setup tests whether external knowledge improves the base multimodal model, and whether MemeLens distillation adds benefit beyond retrieved context in both English and Bengali settings.

4. Validation

Experimental setup. We evaluate all variants on held-out test sets using accuracy, macro precision, macro recall, and macro F1. Macro metrics are used because they weight hateful and non-hateful classes equally. Hyperparameters were selected by a small manual validation search, prioritizing macro F1. FHM uses learning rate $3 \cdot 10^{-5}$, batch size 4, 10 epochs, dropout 0.3, hidden dimension 512, $\lambda_{\text{KD}} = 0.05$, and $\tau = 1.0$. For MUTE fine-tuning, we lower the learning rate to 10^{-5} and increase dropout to 0.5 to reduce overfitting on the smaller Bengali dataset; we also increase λ_{KD} to 0.25, giving the teacher signal more weight.

Dataset	Model	Acc.	Prec.	Rec.	F1
FHM	BLIP-2	64.9	64.9	64.9	64.9
	Context	65.1	65.2	65.1	65.1
	Context + Distill	65.7	65.8	65.7	65.7
MUTE	BLIP-2	70.0	70.3	69.8	70.0
	Context	72.6	69.1	81.8	74.9
	Context + Distill	73.2	71.8	76.7	74.2

Table 1. Student model test metrics (%).

Results and analysis. On FHM, all variants perform similarly: macro F1 increases only from 64.9 for the base model to 65.1 with context and 65.7 with context plus distillation. This weak gain suggests that our retrieval signal does not capture the main difficulty of FHM. Retrieval is based on meme text and a generated caption, but many FHM examples require understanding how a visual cue changes the meaning of otherwise neutral text. If the caption misses this cue, the retrieved knowledge cannot recover it. Thus, FHM likely needs stronger visual grounding or region-level retrieval rather than only text-driven context.

On MUTE, external knowledge has a clearer benefit. Adding Bengali HurtLex context raises macro F1 from 70.0 to 74.9, mainly because false negatives drop from 48 to 29. In other words, the model misses fewer hateful memes. We interpret this as follows: for many MUTE examples, the meme text and generated caption were sufficient to retrieve relevant Bengali hate/offensive cues, and their retrieved lexicons helped the model recognize harmful content. This result is consistent with our hypothesis that language-specific external knowledge is more beneficial in multilingual set-

tings, where culturally grounded expressions and offensive terminology may not be fully captured by pretrained multimodal representations. However, false positives also increase from 47 to 58. This suggests that the context model became more sensitive to hate signals: because Bengali HurtLex is itself a hate/offensive lexicon, retrieved cues may sometimes push the model toward the hateful class more strongly than the full meme context justifies.

Distillation changes the precision–recall trade-off. On FHM, it gives the best scores but only a small gain. On MUTE, context with distillation gives the highest accuracy and precision, and reduces false positives from 58 to 48. However, recall drops compared with context only and macro F1 decreases slightly from 74.9 to 74.2. This suggests that MemeLens supervision regularizes the model and reduces over-prediction, but does not add as much information as targeted Bengali retrieval. A likely reason is that our distillation is chat-based: teacher probabilities are parsed from structured text outputs rather than taken from calibrated logits, making the teacher signal informative but noisy.

Model	Total	Trainable	Preprocess	Forward
Base	993M	6.9M	1.1s	15.2ms
Context	1.00B	17.1M	1.2s	24.3ms
Context + KD	1.00B	17.1M	1.2s	24.3ms
MemeLens	8.77B	–	69ms	1.2s

Table 2. Efficiency comparison. Preprocessing and Forward times are obtained using one A100-40g GPU. Distillation adds no parameters because it only changes the training loss.

Deployability. Table 2 shows that our deployed models are substantially smaller than the MemeLens teacher. Even the context model loads about $8.7\times$ fewer total parameters than MemeLens-VLM, and trains only 17.1M parameters, about $513\times$ fewer than the teacher’s total parameter count. The distillation variant has the same inference cost as the context model, since MemeLens is used only offline to generate soft labels and is removed at test time. Overall, this makes the student model more deployable than MemeLens in terms of model size and forward inference speed: it is approximately $50\times$ faster during forward inference, although preprocessing is about 231 ms slower because captions and knowledge embeddings must be generated before prediction.

Limitations. Our evaluation has four main limitations. First, we report one main run per setting, so variance across random seeds is not fully measured. Second, the datasets are relatively small after balancing, especially MUTE, so a few misclassified examples can noticeably change the reported metrics and make validation-based hyperparameter selection less reliable. Third, retrieval uses only textual signals: the meme text and the generated image caption. It does not retrieve knowledge from specific visual regions, so it may miss harmful cues that depend on particular objects, sym-

bols, or people in the image. Fourth, teacher supervision depends on parsed chat responses, which can be noisy and less calibrated than direct teacher logits.

5. Conclusion

We studied whether external knowledge retrieval and MemeLens distillation can improve a lightweight multimodal hateful meme detector. Context helped only slightly on FHM, likely because text- and caption-based knowledge retrieval missed the implicit and difficult visual-text relations required by many examples. On the easier MUTE dataset, Bengali HurtLex improved recall and macro F1, suggesting that language-specific hate lexicons can help when meme text and captions retrieve relevant community-specific cues. Distillation gave limited gains: it produced the best but only slightly improved FHM scores, improved precision on MUTE, but did not outperform context alone in MUTE macro F1. This may be because teacher probabilities were parsed from chat outputs rather than calibrated logits. Future work should use image-region-grounded retrieval, direct teacher logits, and multiple random seeds for more robust evaluation.

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